

Power Tail Decay Gives the Noncommutative Weak Endpoint in the Correct Kernel Orientation

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Status. Candidate partial resolution, likely valid; human review required. The quantitative tail condition gives weak type $(1, 1)$ for the operator when it is imposed in the input variable, including the source's $\mathcal{M}' \cap \widetilde{\mathcal{M}}$ -valued kernels. The one-sided formula printed in the source is in the opposite orientation and gives the endpoint for the Hilbert adjoint. Under the natural two-sided reading, both endpoints follow.

1 The source question and the orientation issue

Let $(\widetilde{\mathcal{M}}, \widetilde{\tau})$ be a semifinite von Neumann algebra and let $\mathcal{M} \subset \widetilde{\mathcal{M}}$ be a von Neumann subalgebra on which $\widetilde{\tau}$ remains semifinite. Put

$$\mathcal{N} = L_\infty(\mathbb{R}^d) \bar{\otimes} \mathcal{M}, \quad \widetilde{\mathcal{N}} = L_\infty(\mathbb{R}^d) \bar{\otimes} \widetilde{\mathcal{M}}.$$

Their natural tensor traces are denoted $\varphi = \int_{\mathbb{R}^d} dx \otimes \tau$ and $\widetilde{\varphi} = \int_{\mathbb{R}^d} dx \otimes \widetilde{\tau}$. The source studies an $L_2(\mathcal{N}) \rightarrow L_2(\widetilde{\mathcal{N}})$ operator

$$Tf(x) = \int_{\mathbb{R}^d} k(x, y) f(y) dy, \quad k(x, y) \in \mathcal{M}' \cap \widetilde{\mathcal{M}}, \quad (1)$$

with

$$\|k(x, y)\|_{\widetilde{\mathcal{M}}} \leq C_0 |x - y|^{-d}. \quad (2)$$

Its Remark 1.4 asks whether a quantitative integral tail can replace pointwise Lipschitz estimates:

We will also suppose that T verifies the *size condition*, for all $x, y \in \mathbb{R}^n$:

$$\|k(x, y)\|_{\widetilde{\mathcal{M}}} \leq \frac{1}{|x - y|^n}.$$

Remark 1.4. This size condition will never appear throughout the proof. It is however used in remark 1.6. Moreover, it is implied by the smoothness condition provided that k goes to 0 at infinity.

On the contrary, the smoothness condition will be crucial to many computations. A Hörmander type condition:

$$\sup_{x \in \mathbb{R}^n} \int_{|x-y| > 2|x-x'|} |k(x, y) - k(x', y)| dy < C$$

would not suffice to use the ideas of this paper. We would at least need a decay of the following type:

$$\sup_{x \in \mathbb{R}^n} \int_{|x-y| > \alpha|x-x'|} |k(x, y) - k(x', y)| dy < \frac{C}{\alpha^n}, \quad \alpha > 2.$$

But we do not know if it is a strong enough condition since part of the proof of pseudo-localisation relies on pointwise estimates of the kernel.

Define, for all $t > 0$ and $f \in \widetilde{\mathcal{N}}$:

$$\lambda_t(f) = \tau(\{f > t\}).$$

The main theorem we are aiming to prove in this article is the following.

Figure 1: The exact question in Remark 1.4 of [1]. Notice that the displayed difference varies the first argument of k .

For (1), cancellation of a bad input on a cube Q produces

$$k(x, y) - k(x, c_Q),$$

so weak type $(1, 1)$ for T naturally asks for regularity in the *second*, or input, variable. The displayed condition in Figure 1 varies the *first*, or output, variable. The distinction cannot be erased by renaming dummy variables.

We prove a stronger power-tail theorem in the input orientation, extend the later scalar-kernel endpoint argument to the source's commutant-valued setting, and then state exactly what the printed condition implies.

2 Statement of the result

For $\beta > 0$, say that k satisfies the input-tail condition $\text{Tail}_{\text{in}}(A, \beta)$ when, for all $y, z \in \mathbb{R}^d$ and $\alpha > 2$,

$$\int_{|x-y| > \alpha|y-z|} \|k(x, y) - k(x, z)\|_{\widetilde{\mathcal{M}}} dx \leq A\alpha^{-\beta}. \quad (3)$$

The exponent need not equal the dimension.

Theorem 1 (Input-tail endpoint). *Assume (2), $\text{Tail}_{\text{in}}(A, \beta)$ for some $\beta > 0$, and boundedness $T : L_2(\mathcal{N}) \rightarrow L_2(\widetilde{\mathcal{N}})$. Then T is of weak type $(1, 1)$: there is a finite constant*

$$C = C(d, \beta, C_0, A, \|T\|_{2 \rightarrow 2})$$

such that for every $f \in L_1(\mathcal{N})$ and $\lambda > 0$,

$$\lambda \widetilde{\varphi}(\mathbf{1}_{(\lambda, \infty)}(|Tf|)) \leq C\|f\|_{L_1(\mathcal{N})}. \quad (4)$$

In particular, the source's decay α^{-d} is more than enough in the input orientation. A useful slightly more general formulation replaces $\alpha^{-\beta}$ in (3) by a decreasing modulus $\omega(\alpha)$ such that

$$\sum_{j \geq 0} \sqrt{\omega(c_d 2^j)} < \infty. \quad (5)$$

3 Proof intuition

The later paper [3] found a Calderón–Zygmund decomposition that avoids pseudolocalization. Its endpoint theorem asks for a square-integral of the input-variable kernel difference on every remote annulus. The source's tail estimate gives an L_1 bound there. The size estimate gives an L_∞ bound there. Multiplying the two bounds gives the required L_2 estimate, and taking a square root explains why the Dini condition is on $\sqrt{\omega}$.

The published proof is written for scalar kernels and notes a central-valued extension. For the source's more general kernels, every kernel value commutes with the input algebra. This is exactly what the row/column factorization in the proof needs: $|K|^2$ can be replaced by $\|K\|^2 \mathbf{1}$ under the trace, with no change to the Cuculescu decomposition.

4 Tail decay implies annular L_2 –Hörmander regularity

Fix an odd dilation factor D_d large enough that, for a cube Q of side $\ell(Q)$, $y, z \in Q$, and $x \notin D_d Q$, the distances $|x - y|$ and $|x - z|$ are comparable to $|x - c_Q|$. Let

$$E_j(Q) = \{x \notin D_d Q : 2^j D_d \ell(Q) \leq |x - c_Q| < 2^{j+1} D_d \ell(Q)\}, \quad j \geq 0.$$

Lemma 2 (Tail-to-square upgrade). *If (2) and (3) hold, then*

$$\sup_Q \sum_{j \geq 0} \sup_{y \in Q} \left(|E_j(Q)| \int_{E_j(Q)} \|k(x, y) - k(x, c_Q)\|_{\widetilde{\mathcal{M}}}^2 dx \right)^{1/2} < \infty. \quad (6)$$

More precisely, its left side is at most $C_{d,\beta} \sqrt{AC_0}$. The same conclusion holds with a modulus ω satisfying (5).

Proof. Fix Q, j, y , and write $z = c_Q$. On $E_j(Q)$,

$$|x - y| \simeq_d 2^j \ell(Q), \quad |x - z| \simeq_d 2^j \ell(Q), \quad |y - z| \lesssim_d \ell(Q).$$

After increasing D_d once, the annulus lies in the tail region of (3) with $\alpha = c_d 2^j > 2$. Hence

$$\int_{E_j(Q)} \|k(x, y) - k(x, z)\|_{\widetilde{\mathcal{M}}} dx \leq C_d A 2^{-j\beta}. \quad (7)$$

The size condition gives

$$\sup_{x \in E_j(Q)} \|k(x, y) - k(x, z)\|_{\widetilde{\mathcal{M}}} \leq C_d C_0, 2^{-jd} \ell(Q)^{-d}. \quad (8)$$

Since $|E_j(Q)| \leq C_d 2^{jd} \ell(Q)^d$, multiplying (7) and (8) yields

$$|E_j(Q)| \int_{E_j(Q)} \|k(x, y) - k(x, z)\|_{\widetilde{\mathcal{M}}}^2 dx \leq C_d A C_0, 2^{-j\beta}.$$

Its square root is summable because $\beta > 0$. Replacing $2^{-j\beta}$ by $\omega(c_d 2^j)$ proves the last assertion. $\square$

The fixed dilation D_d only changes dimensional constants in the standard exceptional projection. Equivalently, one may retain the dilation used in [3]; size controls the finitely many extra shells, and Lemma 2 controls the remote ones.

5 The commutant-valued endpoint argument

Lemma 3 (Extension of the later endpoint theorem). *Let T have the form (1), be bounded $L_2(\mathcal{N}) \rightarrow L_2(\widetilde{\mathcal{N}})$, and have a kernel in $\mathcal{M}' \cap \widetilde{\mathcal{M}}$ satisfying (6). Then (4) holds.*

Proof. We audit the proof of Theorem A in [3]. At height λ , its new noncommutative Calderón–Zygmund decomposition writes

$$f = g + b_d + b_{\text{off}}$$

using Cuculescu projections in $\mathcal{N}$, and constructs a dilated projection $\zeta \in \mathcal{N}$ with

$$\varphi(\mathbf{1} - \zeta) \lesssim_d \lambda^{-1} \|f\|_1.$$

The good term uses only Chebyshev, the L_2 bound for T , and $\|g\|_2^2 \lesssim_d \lambda \|f\|_1$, so it is unchanged when the range is $\widetilde{\mathcal{N}}$. Terms carrying $\mathbf{1} - \zeta$ are controlled by the same support projection estimate and the fact that $\widetilde{\tau}|_{\mathcal{M}} = \tau$.

For a bad diagonal piece supported on Q , cancellation gives

$$\zeta(x) T b_{d,Q}(x) \zeta(x) = \zeta(x) \int_Q [k(x, y) - k(x, c_Q)] b_{d,Q}(y) dy \zeta(x)$$

outside the dilated cube. The trace-ideal inequality $\|ab\|_1 \leq \|a\|_\infty \|b\|_1$ and the L_1 consequence of (6) give the published diagonal estimate verbatim.

Only the off-diagonal piece uses the scalar nature of the kernel in the printed proof. Write

$$K_Q(x, y) = k(x, y) - k(x, c_Q).$$

The factors $\pi_Q f(y) \pi_Q$ and all Cuculescu projections lie in $\mathcal{M}$, while $K_Q(x, y) \in \mathcal{M}' \cap \widetilde{\mathcal{M}}$. They therefore commute, and positivity gives the exact replacement for the scalar square used in [3]:

$$\tilde{\tau}(|K_Q(x, y)|^2 \pi_Q f(y) \pi_Q) \leq \|K_Q(x, y)\|_{\widetilde{\mathcal{M}}}^2 \tau(\pi_Q f(y) \pi_Q). \quad (9)$$

The row/column Cauchy–Schwarz factorization, shell integration, and (6) now give the same bound as in the scalar proof:

$$\sum_Q \|\zeta T b_{\text{off}, Q} \zeta\|_1 \lesssim_d \mathfrak{H}_2(k) \|f\|_1,$$

where $\mathfrak{H}_2(k)$ denotes the left side of (6). Summing the good, exceptional, diagonal, and off-diagonal distribution estimates proves (4). No step asks that K_Q be central; commutation with $\mathcal{M}$ is sufficient. $\square$

Theorem 1 follows immediately from Lemmas 2 and 3.

6 What the formula printed in the source proves

Define the output-tail condition $\text{Tail}_{\text{out}}(A, \beta)$ by

$$\int_{|x-y| > \alpha|x-x'|} \|k(x, y) - k(x', y)\|_{\widetilde{\mathcal{M}}} dy \leq A\alpha^{-\beta} \quad (\alpha > 2). \quad (10)$$

This is exactly the orientation printed in Figure 1 when $\beta = d$.

Corollary 4 (Adjoint and two-sided conclusions). *Assume the setup of Theorem 1.*

1. If (10) holds, the Hilbert-space adjoint $T^\dagger : L_2(\widetilde{\mathcal{N}}) \rightarrow L_2(\mathcal{N})$, restricted to $L_1(\mathcal{N})$, is weak type $(1, 1)$.
2. If both (3) and (10) hold, then T and $T^\dagger|_{L_1(\mathcal{N})}$ are weak type $(1, 1)$.
3. When $\mathcal{M} = \widetilde{\mathcal{M}}$, $T^\dagger$ is the usual integral adjoint with kernel $k^\dagger(u, v) = k(v, u)^*$.

Proof. Because the ambient trace is tracial, there is a trace-preserving conditional expectation $E_{\mathcal{M}} : \widetilde{\mathcal{M}} \rightarrow \mathcal{M}$. On separated supports the restriction of the Hilbert adjoint has kernel

$$h(u, v) = E_{\mathcal{M}}(k(v, u)^*).$$

Conditional expectation is contractive in operator norm, and bimodularity shows that $h(u, v) \in \mathcal{M} \cap \mathcal{M}' = \mathcal{Z}(\mathcal{M})$. Condition (10) for k is condition (3) for h , with the same constants. Also $\|T^\dagger\|_{2 \rightarrow 2} = \|T\|_{2 \rightarrow 2}$. Apply Theorem 1 to the adjoint. The second assertion combines this with Theorem 1 for T , and the third is immediate. $\square$

7 Scope, verification, and literature relation

The result answers the weak-endpoint question affirmatively under the natural two-sided replacement of the source’s preceding two Lipschitz estimates. It also shows that *any* positive power tail, not only α^{-d} , is sufficient in the input orientation. It does not prove that the original pseudolocalization estimate survives: the later decomposition was designed to avoid that estimate.

The literal one-sided formula (10) only supplies the endpoint for the adjoint. Weak $(1, 1)$ for $T^\dagger$ plus the L_2 bound gives strong bounds for T above 2 by interpolation and duality, but it does not

reach weak L_1 for T . A proof-grade counterexample to the remaining one-sided implication was not found; no such implication is claimed here.

Hong–Lai–Xu [2] prove an endpoint under a summable annular L_2 condition for scalar kernels tensored with the identity. Cadilhac, Conde-Alonso, and Parcet [3] give the decomposition used above and an annular L_2 –Hörmander theorem, noting extensions to arbitrary noncommutative input spaces and central-valued kernels. The new deductions here are the square-Dini tail criterion, the explicit orientation correction, and the commutant-valued range extension matching [1]. Novelty is provisional; eight distinct proof and counterexample upgrades are recorded in the accompanying attempt log.

References

- [1] L. Cadilhac, *Weak boundedness of Calderón–Zygmund operators on noncommutative L_1 -spaces*, J. Funct. Anal. 274 (2018), 769–796; arXiv:1702.06536.
- [2] G. Hong, X. Lai, and B. Xu, *Maximal singular integral operators acting on noncommutative L_p -spaces*, arXiv:2009.03827.
- [3] L. Cadilhac, J. M. Conde-Alonso, and J. Parcet, *Spectral multipliers in group algebras and noncommutative Calderón–Zygmund theory*, arXiv:2105.05036.